Design and Model Summary Report

Design Summary

Summary of the design structure, including the total number of runs, the amount of replication, the number of center points, and the composition of continuous and categorical factors.

Design	Runs	Replicates	Center Points	Continuous	Categorical
Central Composite	17	2	3	3	0

Factor Summary

Summary of each factor included in the design, showing its type, number of levels, actual level values, and corresponding coded levels.

Factor	Type	Levels Count	Levels	Coded Levels
solv_r	cont	3	[4.0, 5.0, 6.0]	[-1. 0. 1.]
temp	cont	3	[35.0, 40.0, 45.0]	[-1. 0. 1.]
base	cont	3	[1.2, 1.4, 1.6]	[-1. 0. 1.]

Regression Summary

Model Terms

The prediction model for conv, prod, UK2 is built with the following terms:

- Intercept

- Linear Terms: solv_r, temp, base

- 2-Term Interactions: solv_r × temp, solv_r × base, temp × base

- Quadratic Terms: solv_r^2, temp^2, base^2

- 3-Term Interactions:

Variance Inflation Factors (VIF)

Variance Inflation Factors (VIFs) quantify the degree of multicollinearity among the regression terms in the fitted model.

- VIF = 1: No multicollinearity, predictors are perfectly orthogonal (Ideal).

- 1 < VIF < 5: Low to moderate multicollinearity (Acceptable).

- VIF > 5: Potential concerns, predictors may be correlated.

- VIF > 10: Serious multicollinearity, coefficients might be unstable (Problematic).

Term	VIF	Term	VIF
solv_r	1.0	temp:base	1.0
temp	1.0	solv_r^2	1.537
base	1.0	temp^2	1.537
solv_r:temp	1.0	base^2	1.537
solv_r:base	1.0		

Leverages

Leverage values for each design point, indicating their influence on the model fit.

Run	Leverage	Run	Leverage
1	0.7947	13	0.5155
2	0.7947	14	0.5155
3	0.7947	15	0.1831
4	0.7947	16	0.1831
5	0.7947	17	0.1831
6	0.7947		
7	0.7947		
8	0.7947		
9	0.5155		
10	0.5155		
11	0.5155		
12	0.5155		

Response Conditions

Summary of constraints and optimization goals applied to each response during the design of experiments.

Response	Lower Limit	Upper Limit	Goal
conv	No	No	Maximize
prod	No	No	Maximize
UK2	No	No	Maximize

Model Summary for Response: conv

Summary of the fitted MLR model for conv, presenting the regression coefficients, ANOVA results, model and lack-of-fit F-tests, goodness-of-fit statistics, replicate analysis, and predicted responses.

Coefficient Summary

Term	Coefficient	Std. Error	Conf. Int.	Lower CI	Upper CI	p-value
Int	97.1451	0.4795	1.1337	96.0113	98.2788	0.0
solv_r	0.82	0.3543	0.8379	-0.0179	1.6579	0.0539
temp	3.7	0.3543	0.8379	2.8621	4.5379	0.0
base	8.31	0.3543	0.8379	7.4721	9.1479	0.0
solv_r:temp	0.225	0.3962	0.9368	-0.7118	1.1618	0.5878
solv_r:base	-0.15	0.3962	0.9368	-1.0868	0.7868	0.7162
temp:base	-2.6	0.3962	0.9368	-3.5368	-1.6632	0.0003
solv_r^2	-0.5789	0.6845	1.6187	-2.1976	1.0398	0.4257
temp^2	-0.8789	0.6845	1.6187	-2.4976	0.7398	0.24
base^2	-5.5289	0.6845	1.6187	-7.1476	-3.9102	0.0001

Model F-test

Overall significance of the regression model assessed via the F-test.

F-value	F-crit (95%)	F-crit (99%)	p-value
93.4172	3.6767	6.7188	0.0

Lack-of-Fit F-test

Assessment of model adequacy through the Lack-of-Fit F-test.

F-value	F-crit (95%)	F-crit (99%)	p-value
175.3704	19.2964	99.2993	0.0057

Model Metrics

Key performance metrics for the regression model.

R^2	Adj. R^2	Q^2	Adj Q^2	RMSE	RMSE (CV)
0.9917	0.9811	0.8941	0.9003	1.1205	2.5755

ANOVA Table

Analysis of variance (ANOVA) table summarizing the sources of variation in the model.

Source	SS	dof	MS
Total	1064.359	16	66.522
Regression	1055.57	9	117.286
Residuals	8.789	7	1.256
Pure Error	0.02	2	0.01
Lack of Fit	8.769	5	1.754

Replicate Summary

Summary statistics for replicate runs in the design.

Run Index	n*	Mean	Variance	Std. Dev.	dof
(14, 15, 16)	3	97.5	0.01	0.1	2

Predicted Responses

Predicted response values from the fitted model for each design point.

Run	conv	Pred. conv
1	75.7	74.8
2	96.7	96.9
3	86.8	87.0
4	99.8	98.7
5	75.3	76.3
6	98.1	97.8
7	89.7	89.3
8	99.7	100.5
9	98.5	97.4
10	94.1	95.7
11	99.4	100.0
12	92.6	92.6
13	99.5	99.9
14	83.2	83.3
15	97.5	97.1
16	97.4	97.1
17	97.6	97.1

Model Summary for Response: prod

Summary of the fitted MLR model for prod, presenting the regression coefficients, ANOVA results, model and lack-of-fit F-tests, goodness-of-fit statistics, replicate analysis, and predicted responses.

Coefficient Summary

Term	Coefficient	Std. Error	Conf. Int.	Lower CI	Upper CI	p-value
Int	90.2462	0.5522	1.3058	88.9404	91.552	0.0
solv_r	0.371	0.4081	0.965	-0.594	1.336	0.3935
temp	1.486	0.4081	0.965	0.521	2.451	0.0083
base	3.934	0.4081	0.965	2.969	4.899	0.0
solv_r:temp	0.0125	0.4563	1.0789	-1.0664	1.0914	0.9789
solv_r:base	-0.365	0.4563	1.0789	-1.4439	0.7139	0.45
temp:base	-4.54	0.4563	1.0789	-5.6189	-3.4611	0.0
solv_r^2	-0.5958	0.7885	1.8644	-2.4602	1.2686	0.4745
temp^2	-0.4308	0.7885	1.8644	-2.2952	1.4336	0.6017
base^2	-6.7408	0.7885	1.8644	-8.6052	-4.8764	0.0001

Model F-test

Overall significance of the regression model assessed via the F-test.

F-value	F-crit (95%)	F-crit (99%)	p-value
37.6184	3.6767	6.7188	0.0

Lack-of-Fit F-test

Assessment of model adequacy through the Lack-of-Fit F-test.

F-value	F-crit (95%)	F-crit (99%)	p-value
14.7187	19.2964	99.2993	0.0648

Model Metrics

Key performance metrics for the regression model.

R^2	Adj. R^2	Q^2	Adj Q^2	RMSE	RMSE (CV)
0.9797	0.9537	0.688	0.7064	1.2906	3.25

ANOVA Table

Analysis of variance (ANOVA) table summarizing the sources of variation in the model.

Source	SS	dof	MS
Total	575.568	16	35.973
Regression	563.908	9	62.656
Residuals	11.659	7	1.666
Pure Error	0.308	2	0.154
Lack of Fit	11.351	5	2.27

Replicate Summary

Summary statistics for replicate runs in the design.

Run Index	n*	Mean	Variance	Std. Dev.	dof
(14, 15, 16)	3	90.6667	0.1542	0.3927	2

Predicted Responses

Predicted response values from the fitted model for each design point.

Run	prod	Pred. prod
1	72.79	71.8
2	89.06	89.47
3	83.24	83.82
4	84.78	83.34
5	71.96	73.24
6	90.2	89.46
7	85.89	85.32
8	82.54	83.38
9	90.83	90.02
10	87.84	89.28
11	90.71	91.3
12	88.29	88.33
13	86.51	87.44
14	79.87	79.57
15	90.45	90.25
16	91.12	90.25
17	90.43	90.25

Model Summary for Response: UK2

Summary of the fitted MLR model for UK2, presenting the regression coefficients, ANOVA results, model and lack-of-fit F-tests, goodness-of-fit statistics, replicate analysis, and predicted responses.

Coefficient Summary

Term	Coefficient	Std. Error	Conf. Int.	Lower CI	Upper CI	p-value
Int	2.6465	0.1278	0.3023	2.3442	2.9487	0.0
solv_r	0.266	0.0945	0.2234	0.0426	0.4894	0.0259
temp	0.91	0.0945	0.2234	0.6866	1.1334	0.0
base	2.378	0.0945	0.2234	2.1546	2.6014	0.0
solv_r:temp	0.1238	0.1056	0.2497	-0.126	0.3735	0.2797
solv_r:base	0.1112	0.1056	0.2497	-0.1385	0.361	0.3272
temp:base	0.7513	0.1056	0.2497	0.5015	1.001	0.0002
solv_r^2	-0.2663	0.1825	0.4316	-0.6979	0.1652	0.1878
temp^2	-0.1463	0.1825	0.4316	-0.5779	0.2852	0.449
base^2	0.7737	0.1825	0.4316	0.3421	1.2052	0.0038

Model F-test

Overall significance of the regression model assessed via the F-test.

F-value	F-crit (95%)	F-crit (99%)	p-value
89.5609	3.6767	6.7188	0.0

Lack-of-Fit F-test

Assessment of model adequacy through the Lack-of-Fit F-test.

F-value	F-crit (95%)	F-crit (99%)	p-value
3.5495	19.2964	99.2993	0.2343

Model Metrics

Key performance metrics for the regression model.

R^2	Adj. R^2	Q^2	Adj Q^2	RMSE	RMSE (CV)
0.9914	0.9803	0.9236	0.9281	0.2987	0.5711

ANOVA Table

Analysis of variance (ANOVA) table summarizing the sources of variation in the model.

Source	SS	dof	MS
Total	72.556	16	4.535
Regression	71.932	9	7.992
Residuals	0.625	7	0.089
Pure Error	0.063	2	0.032
Lack of Fit	0.561	5	0.112

Replicate Summary

Summary statistics for replicate runs in the design.

Run Index	n*	Mean	Variance	Std. Dev.	dof
(14, 15, 16)	3	2.6533	0.0316	0.1779	2

Predicted Responses

Predicted response values from the fitted model for each design point.

Run	UK2	Pred. UK2
1	0.47	0.44
2	3.54	3.47
3	0.78	0.51
4	6.52	6.55
5	0.53	0.5
6	3.71	3.98
7	1.0	1.07
8	7.52	7.55
9	2.98	2.65
10	1.77	2.11
11	3.26	3.41
12	1.73	1.59
13	6.05	5.8
14	0.78	1.04
15	2.81	2.65
16	2.46	2.65
17	2.69	2.65